Singapore Institute of Technology - University of Glasgow
Joint Degree in Computing Science Degree Programme

CSC3101 Capstone Project

Please complete the following form and attach it to the Capstone Report submitted.

Capstone Period: **31 08 2026** **to** **08 04 2027**
Assessment Trimester: **Start Trimester**
Project Type: **Industry**

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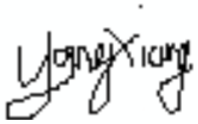
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I hereby acknowledge that I have engaged and discussed with my **Academic Supervisor** and **Work Supervisor** on the contents of this Capstone Report (Problem Definition) and have sought approval to release the report to the Singapore Institute of Technology and the University of Glasgow.

Signature 

Date: 28 09 2026

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Singapore Institute of Technology - University of Glasgow
Joint Degree in Computing Science Degree Programme

Capstone Report (Problem Definition)
"Automated VM Provisioning and Integrated Monitoring on Proxmox VE: An Infrastructure-as-Code Pipeline"

For **Start Trimester** from 31 08 2026 to 08 04 2027

Hooi Yong Xiang
Student ID: 2402752

Work Supervisor: *Wong King Teck*

Academic Supervisor: *Bogdan Cautis*

Submitted as part of the requirement for CSC3101 Capstone Project

1. Introduction

Enterprise IT is shifting toward declarative automation and continuous deployment. Yet in hybrid and private clouds, server administrators still face friction when managing virtual infrastructure lifecycles manually [1]. The goal is clear: full Infrastructure-as-Code (IaC) maturity. Virtual host provisioning and telemetry configuration should run as one automated pipeline. This capstone project designs, implements, and evaluates an automated IaC pipeline for rapid provisioning and integrated monitoring of virtual infrastructure. Open source integration such as proxmox VE and Zabbix, offers a secondary benefit by reducing reliance on proprietary platforms. The primary contribution is a replicable, fully automated end-to-end workflow that minimizes human intervention, thereby eliminating configuration errors and improving compute monitoring coverage.

1.1 Problem Definition

1.1.1 High Provisioning Latency and Excessive Manual Overhead

Traditional server administration is manual. Deploying a new virtual machine (VM) means navigating hypervisor graphical user interfaces (GUIs), configuring compute and storage, installing an operating system, and assigning network parameters. A 2024 MSc study measured manual deployment of a simple AWS serverless application at approximately 30 minutes, with variance depending on the administrator's familiarity with the console and services involved. The same study found that IaC tools reduced deployment time by over 80% [2]. As virtual environments scale, manual GUI work becomes a severe bottleneck. It slows deployment and hinders rapid infrastructure scaling.

1.1.2 Configuration Errors and Infrastructure Drift

Manual provisioning invites human error and inconsistency. Repetitive setup across multiple hosts often creates subtle discrepancies such as misconfigured services, mismatched security policies, or unpatched base packages [3], [4]. Over time, these variations cause configuration drift. Drift makes environments difficult to audit, reproduce, or troubleshoot. It also increases vulnerability to operational outages [4].

1.1.3 Disconnected Telemetry Onboarding and Operational Blind Spots

In many organizations, virtualization management and compute monitoring are decoupled. Once a VM is provisioned, the administrator must separately log into the monitoring platform, install a telemetry agent, and finally manually map monitoring templates to the host. These tasks are non-atomic and depend on administrative discipline. Resulting in newly deployed VMs are often forgotten or delayed during onboarding and run without active telemetry [4]. Such blind spots prevent timely incident detection, performance bottleneck analysis, and proactive SLA enforcement [5].

2. Project Objectives

The main goal of this capstone project is to build an end-to-end automated IaC pipeline that unifies virtual machine provisioning with automated monitoring telemetry.

1. **Deliver an Automated IaC Pipeline**: Architect and deploy an integrated pipeline using OpenTofu (bpg/proxmox provider), Ansible, Cloud-Init, and Zabbix API to manage the full lifecycle from VM request to active monitoring.

2. **Reduce Provisioning Deployment Time**: Reduce total VM deployment and onboarding cycle time from a manual baseline of 45 to 60 minutes per VM to under 10 minutes from trigger to active Zabbix monitoring (>80% reduction).

3. **Automate Post-Provisioning Configuration**: Achieve 100% automated post-boot OS configuration and agent installation for Linux VMs in the test matrix via Ansible playbooks and Cloud-Init, requiring zero manual terminal or GUI intervention post-trigger.

4. **Target 100% Monitoring Coverage**: Target 100% monitoring coverage for successfully provisioned VMs under test conditions, verified by automated Zabbix host creation and template linking checks.

5. **Establish Repeatable Infrastructure Standards**: Eliminate configuration drift by enforcing version-controlled declarative templates and Ansible playbooks for all deployed virtual machines.

3. Scope and Limitations

The scope of this project encompasses the development of an automated orchestration pipeline using OpenTofu (leveraging the bpg/proxmox provider), Ansible (for post-boot configuration management), Cloud-Init (for automated OS bootstrap), and Zabbix API integration (for immediate host auto-registration). This conceptual architecture and component interactions is illustrated in **Figure 1**.

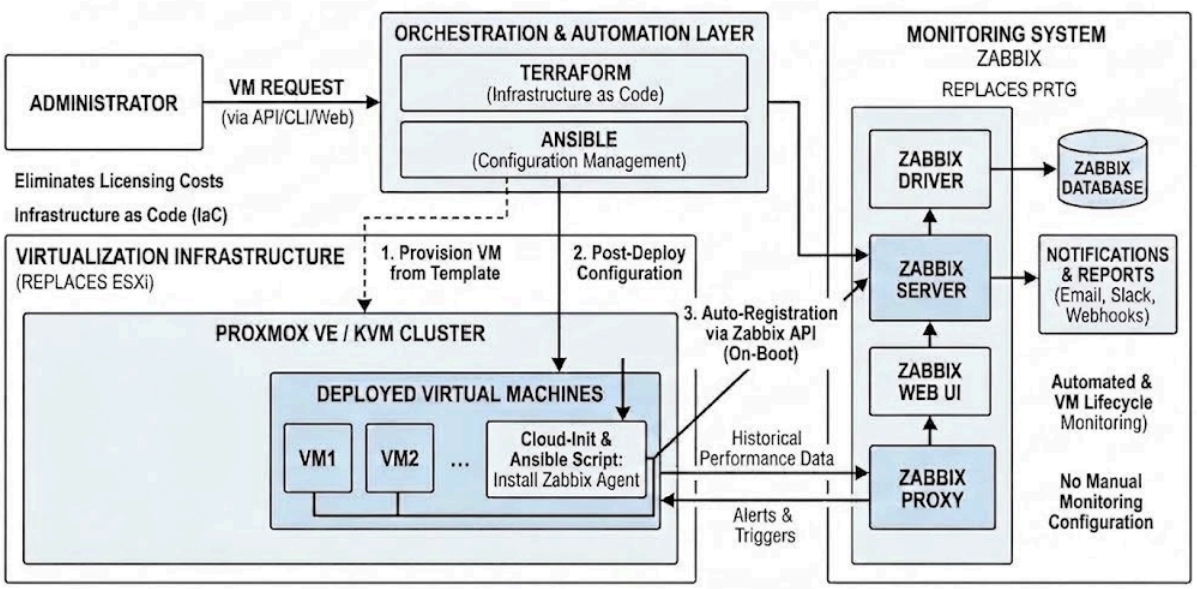


Figure 1. Architectural overview of the end-to-end infrastructure-as-Code provisioning and automated Zabbix monitoring pipeline

- **Target Operating Systems**: Provisioning workflows are strictly scoped to Linux enterprise environments, specifically Ubuntu Server LTS and Alma Linux. Windows Server with Cloudbase-Init is an optional stretch goal, dependent on remaining schedule.

- **Image Management**: Base templates are instantiated from official cloud-init images, configured once manually, and version-controlled as static baselines.

- **Execution & Secrets Management**: Upon Terraform infrastructure provisioning, a post-creation execution step triggers Ansible playbooks against a dynamic inventory generated from Terraform outputs. All sensitive parameters, including Zabbix API tokens and Proxmox credentials, are managed securely via environment variables and Ansible Vault with least-privilege tokens and TLS verification.

- **Testing Environment & Exclusions**: The pipeline will be evaluated within an isolated single-datacenter lab host. Multi-region public cloud orchestrations and container platforms (e.g., Kubernetes) are outside the scope of this project.

4. Evaluation Methodology

To rigorously evaluate the pipeline against the baseline manual workflow, the project will utilize the following framework:

No.	Metric	Benchmark
1	Test Cases	Deployment matrix consisting of 10 Linux VMs (5 Ubuntu Server LTS, 5 Alma Linux instances).
2	Target Metrics	Total request-to-ready time, total request-to-monitored time, provisioning success rate, configuration idempotency, and drift detection capabilities.
3	Comparison	Quantitative benchmark comparing the manual GUI deployment baseline against the automated IaC pipeline.
4	Resilience Testing	Pipeline error handling under failure conditions, including execution reruns, invalid Zabbix API token handling, and simulated network delays.

5. References (Heading 1)

[1] M. Artac, T. Borovssak, E. Di Nitto, M. Guerriero, and D. A. Tamburri, 'DevOps: Introducing Infrastructure-as-Code', IEEE Xplore. Accessed: Sep. 28, 2026. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/7965401

[2] H. Suresh, 'IaC for Secure Serverless: A Comprehensive Approach to Deployment and Configuration Management MSc Research Project Programme Name', Dec. 2024. Accessed: Sep. 28, 2026. [Online]. Available: https://norma.ncirl.ie/8158/1/hrishinpadikkaparambilsuresh.pdf

[3] A. Rahman, R. Mahdavi-Hezaveh, and L. Williams, 'A Systematic Mapping Study of Infrastructure as Code Research', *Information and Software Technology*, vol. 108, pp. 65–77, Apr. 2019, doi: 10.1016/j.infsof.2018.12.004.

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[5] M. Usman, S. Ferlin, A. Brunstrom, and J. Taheri, 'A Survey on Observability of Distributed Edge & Container-Based Microservices', *IEEE Access*, vol. 10, pp. 1–1, 2022, doi: 10.1109/access.2022.3193102.

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